ECE459: Programming for Performance Winter 2027

Lecture 29 — Frontend Performance

Jeff Zarnett *2026-11-10*

Feelings and Facts

Most topics of the course focus on how to actually improve the execution speed of the work to be done, which could be measured in either CPU time or wall clock time reductions (or, even better, both!). Frontend performance is unlike those kinds of improvements, because the user's feelings matter. If they *perceive* the page as fast, that's as good as, or better than, the page actually being faster.

Broadly, three main things help the user perceive the page as fast. Acknowledgement of their actions is a big one: showing the user that their click has done something right away leads to a lot less frustration. You might have heard me [JZ] tell the story about how slow acknowledgement of the purchase action caused me to buy two train tickets when I only needed one. The second thing users want is an indication of progress: show a loading bar, a spinner, or some other sign that makes it clear that the thing they asked for is being worked on (or at least in a queue). The last thing is the most controversial: make the user think it's done when it's not. Work that is completed after the user thinks the page is finished loading is "free".

Let's talk about the thresholds that matter to users:

1. Under 0.1 seconds is "instant".
2. Between 1 and 10 seconds, the delay is noticeable, so we need some sort of loading spinner.
3. After 10 seconds, the user is impatient, so we need to try to find a way to end the waiting before the work is done. If we can't do that, then don't keep the user waiting: tell them we'll notify them when it's ready, or send them the result when it's done.

Motivation. The user experience matters here. Aside from the correctness issues like buying two train tickets, there's also the risk that customers just... stop using your app or abandon whatever they were doing. The absence of complaints about responsiveness is noted in the surveys, including a source mentioned earlier [KSNH15]. We could interpret this positively, as in, performance is not a major concern. Or, users who give up don't bother to write reviews.

If we look at a study on video delays [KS12], users start to abandon a video load if it does not start within two seconds. They also cite sources saying users abandon a website if the page does not load in 4 seconds and that even a delay of 400 ms is correlated with lower usage. Faster internet speeds make this problem worse, because users accustomed to speed will not wait for slow pages, though their willingness to wait is moderated by their device: mobile network waiting is tolerated better than fibre connections. What the user is waiting for may also play a role: if the data needed is small, the user's patience is small. If I just want to get the train departure times for the Kitchener GO line from Union Station in Toronto, I might be in a big hurry, but even if I'm not, this is a small amount of data and should take practically no time to deliver. If I asked for an analysis of all engineering students' performance on midterms in the last five years, yes, I can wait a little longer.

App store reviews being insufficient to motivate this, let the money do the talking: Deloitte Digital and 55 showed that a 0.1 second improvement in 15 retail brands' mobile website resulted in 8.4% more conversions and 9.2% higher order value [D20]. Actually, the whole source document goes over the impact of speed improvements on purchase behaviour across different kinds of products: retail, travel, luxury purchases, but also things like lead generation.

Trickery of Timing

If the goal is to make the user feel as though the work is done sooner, what has to happen? We can't just lie and say "it's totally done, you just can't see it yet!" like an eight-year-old who forgot to do his math homework. We'll also exclude from discussion things that look like only loading the first n rows and then loading the rest when the user scrolls down. Incremental or paginated loading is a good idea, of course, but it does fall firmly under "do less work" so that can be done independently of the ideas in this topic.

Before we think about anything more exotic, let's look at showing a partial page as soon as we can. There are levels to this – we can show a page that just says things are loading, or show partial results (or both). Let's start with loading the shell. It's the frame around where the content will go, while we wait for the content to be delivered.

A couple quick definitions from Google's `web.dev` page. The *first contentful paint (FCP)* is the time it takes before any part of the page's content is shown on the screen (this can include text, images, backgrounds, etc). The *largest contentful paint (LCP)* is the time when the biggest element (usually the main content) is drawn on the screen. Unless a page is super simple, these are not going to be the same: there will be some user-visible time between the first and largest contentful paints.

FCP and LCP are measured as percentiles and people may choose the 75th or 90th percentile times for page loads in goal-setting. This helps even out some of the variability that naturally results in computers and accounts for things like some customers, cohorts, or train schedules are larger or smaller than others. Faster is better for both of these, though we can see from earlier that 4 seconds seems to be a limit of user patience and 4.0 seconds is the threshold of "poor" performance in the eyes of Google. FCP targets look more like 1.8 seconds.

References

[D20] Deloitte Digital and 55. Milliseconds make millions: A study on how improvements in mobile site speed positively affect a brand's bottom line. Technical report, Deloitte Ireland LLP, 2020. Commissioned by Google. URL: `https://www.thinkwithgoogle.com/_qs/documents/9757/Milliseconds_Make_Millions_report_hQYAbZJ.pdf`.

[KS12] S. Shunmuga Krishnan and Ramesh K. Sitaraman. Video stream quality impacts viewer behavior: Inferring causality using quasi-experimental designs. In *Proceedings of the 2012 Internet Measurement Conference (IMC '12)*, 2012. `doi:10.1145/2398776.2398799`.

[KSNH15] Hammad Khalid, Emad Shihab, Meiyappan Nagappan, and Ahmed E. Hassan. What do mobile app users complain about? *IEEE Software*, 32(3):70–77, 2015. `doi:10.1109/MS.2014.50`.